

MSK Pharm Drug Chart

Every drug covered in the three MSK pharmacology lectures, with its mechanism, uses, key toxicities and the one pearl most likely to be tested. Yellow column = high yield.

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Treatment of Osteoporosis
DMARDs & Biologics
Inflammation Conference
STEP 1 added board fact beyond the lecture

TREATMENT OF OSTEOPOROSIS

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Calcium

Vitamin D

Bisphosphonates

Denosumab

Raloxifene

Estrogens

Conjugated estrogen / bazedoxifene

Calcitonin

Teriparatide

Abaloparatide

Romosozumab

DMARDS & BIOLOGICS

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Methotrexate

Hydroxychloroquine

Sulfasalazine

Azathioprine

Minocycline

Leflunomide

Mycophenolate mofetil

TNF-α inhibitors

Anakinra

Rituximab

Abatacept

Tocilizumab

Belimumab

Tofacitinib

IL-17 and IL-12/23 inhibitors

IVIG

Pegloticase

INFLAMMATION CONFERENCE

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NSAIDs

Glucocorticoids

Cyclophosphamide

Calcineurin inhibitors

Anifrolumab

ALSO USED IN INFLAMMATION CONFERENCE CASES

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Methotrexate

Hydroxychloroquine

Azathioprine

Leflunomide

Mycophenolate mofetil

TNF-α inhibitors

Rituximab

Belimumab

Treatment of Osteoporosis

Classify by mechanism · indications, contraindications, adverse effects · sequencing

DRUG	MECHANISM	USES	TOXICITIES & CONTRAINDICATIONS	HIGH-YIELD PEARL
FOUNDATION – CORRECT IN EVERYONE FIRST				
Calcium 1,200 mg/day total (diet first)	Substrate for hydroxyapatite; adequate intake suppresses PTH-driven resorption.	Adjunct to every osteoporosis drug.	Constipation (carbonate). Caution: hypercalcemia, calcium stones.	Correct calcium and vitamin D before any antiresorptive or romosozumab — those drugs can precipitate hypocalcemia.
Vitamin D ≥1,000 IU/day	↑ Intestinal Ca ²⁺ and phosphate absorption; needed for mineralization.	Adjunct; replete deficiency before therapy.	Excess → hypercalcemia.	Pairs with calcium: deficiency must be fixed first to avoid drug-induced hypocalcemia.
ANTIRESORPTIVES				

DRUG	MECHANISM	USES	TOXICITIES & CONTRAINDICATIONS	HIGH-YIELD PEARL
Bisphosphonates Alendronate, risedronate (PO) · ibandronate (PO/IV) · zoledronic acid (IV yearly)	Pyrophosphate analogs bind hydroxyapatite → taken up by osteoclasts. N-containing BPs inhibit farnesyl pyrophosphate synthase (mevalonate pathway) → osteoclast apoptosis.	First-line for postmenopausal, male and glucocorticoid-induced osteoporosis. Also hypercalcemia , Paget disease , bone metastases, osteogenesis imperfecta.	Oral: pill esophagitis . IV: acute-phase flu-like reaction. Rare: ONJ , atypical femoral fracture , hypocalcemia. CI : CrCl <30–35, hypocalcemia, esophageal disease / can't stay upright.	Take fasting with plain water, upright 30–60 min . Only class allowed a drug holiday (reassess at 3–5 yr). Ibandronate = vertebral-only → not first line . Cheap/generic → first choice when uninsured.
Denosumab Anti-RANKL mAb · SC every 6 months	Fully human monoclonal antibody to RANKL → blocks RANKL–RANK → ↓ osteoclast formation, function, survival.	High-risk PMO and men; GIO; bone loss on androgen deprivation (prostate cancer) or aromatase inhibitors (breast cancer).	Hypocalcemia , eczema/dermatitis, cellulitis; rare ONJ/AFF. CI : hypocalcemia, pregnancy.	Boxed warning: severe hypocalcemia in advanced CKD / dialysis (very high yield) — even though it is not renally cleared. No drug holiday : stopping → rebound resorption and multiple vertebral fractures → switch to a bisphosphonate.
Raloxifene SERM · PO daily	Estrogen agonist in bone , antagonist in breast , neutral in uterus.	PMO prevention/treatment in younger women with breast-cancer risk. ↓ vertebral fracture only; ↓ ER+ breast cancer 50–75%.	VTE , hot flashes, leg cramps. CI : VTE history.	VTE is the tested risk. No hip benefit — skip after a hip fracture. No ↑ uterine cancer (unlike tamoxifen).
Estrogens ± progestin if uterus intact	↓ RANKL, ↑ OPG → fewer, less active osteoclasts; ↑ osteoblast collagen; ↑ GI calcium absorption.	Prevention of PMO — best in early menopause with vasomotor symptoms.	VTE, breast tenderness, bleeding, ↑ BP, ↑ triglycerides. CI : breast/estrogen-dependent cancer, VTE/stroke/MI, migraine with aura, undiagnosed bleeding, pregnancy.	Estrogen loss is the main driver of postmenopausal bone loss. Stopping it → rapid BMD loss → follow with a bisphosphonate . WHI: ↑ MI, stroke, breast cancer, VTE; ↓ hip fracture.
Conjugated estrogen / bazedoxifene Tissue-selective estrogen complex (TSEC)	Estrogen for bone and hot flashes + bazedoxifene (a SERM) that shields the endometrium and breast.	PMO prevention with vasomotor symptoms in women with a uterus .	As for estrogen; same contraindications.	Endometrial protection without a progestin .
Calcitonin Thyroid C-cell hormone	Directly inhibits osteoclasts.	Listed among antiresorptives; weak, vertebral-only benefit.	—	Not first line. Reduces vertebral fractures only (with raloxifene and ibandronate).
ANABOLICS — BUILD BONE				
Teriparatide PTH(1-34) · SC daily	Intermittent PTH-receptor stimulation → ↓ osteoblast apoptosis, ↑ differentiation → net bone formation . (Continuous PTH → resorption.)	High/very-high-risk PMO, men, GIO. Beat risedronate (VERO).	Hypercalcemia , hypercalciuria/stones, orthostatic hypotension, dizziness, nausea.	Avoid with ↑ osteosarcoma risk: Paget disease, prior skeletal radiation, unexplained ↑ ALP, open epiphyses, bone metastases. Must be followed by an antiresorptive . Choose it over romosozumab after a recent MI/stroke.
Abaloparatide PTHrP analog · SC daily, ≤2 yr	Binds PTH1R in the R0 conformation → maximizes formation with less resorption.	PMO (and men); follow with alendronate.	Orthostatic hypotension, dizziness, palpitations; less hypercalcemia than teriparatide.	Same osteosarcoma-risk cautions as teriparatide. R0-selective = more anabolic, less pro-resorptive .

DRUG	MECHANISM	USES	TOXICITIES & CONTRAINDICATIONS	HIGH-YIELD PEARL
Romosozumab Anti-sclerostin mAb · SC monthly × 12	Blocks sclerostin (osteocyte Wnt inhibitor) → dual action : ↑ formation and ↓ resorption.	PMO at high fracture risk. Beat alendronate (ARCH); raises BMD more than bisphosphonates.	Arthralgia, headache; rare ONJ/AFF. CI : hypocalcemia.	Boxed warning: MI, stroke, CV death — do not start within 1 year of MI or stroke. Only dual-action agent; 12 doses, then an antiresorptive.

DMARDs & Biologics

Why DMARDs/biologics · treat to target · anti-drug antibodies · key toxicities

DRUG	MECHANISM	USES	TOXICITIES & CONTRAINDICATIONS	HIGH-YIELD PEARL
CONVENTIONAL DMARDs				
Methotrexate Low dose, weekly PO/SC + folic acid also in Inflammation Conf.	Inhibits dihydrofolate reductase (antimetabolite); at low dose also ↑ adenosine (anti-inflammatory).	Cornerstone RA drug ; PsA, sarcoid, DM steroid-sparing; lupus joints/skin (not kidneys); GPA maintenance if GFR ≥50.	↑ LFTs, cytopenias , early lung toxicity, teratogenic. CI : renal insufficiency (levels build fast). Monitor CBC + liver enzymes.	Lowest discontinuation rate of the DMARDs. Leucovorin rescues toxicity. Lowers anti-drug antibodies when added to a biologic.
Hydroxychloroquine Antimalarial also in Inflammation Conf.	↑ Lysosomal pH, ↓ TLR signaling → ↓ antigen presentation by macrophages.	Standard in lupus (first therapy for mild SLE; kept in nearly all); RA in combination.	Retinal toxicity → periodic eye exams.	Mild lupus (rash, ulcers, arthralgia, normal kidneys) → hydroxychloroquine .
Sulfasalazine	Sulfa-containing DMARD (sulfapyridine + 5-ASA).	RA, usually in combination; moderately helpful alone.	Sulfa allergy ; GI and CNS (headache, fever); rash, ↑ LFTs, leukopenia, agranulocytosis .	Think sulfa allergy and marrow suppression.
Azathioprine Prodrug of 6-mercaptopurine also in Inflammation Conf.	Purine analog → 6-MP/6-TG incorporated into DNA; targets lymphocytes.	Maintenance after cyclophosphamide (lupus nephritis, vasculitis); GPA maintenance if GFR <50; dropped for RA.	Marrow suppression , GI, viral infections, malignancy.	+ allopurinol → 6-MP toxicity (xanthine oxidase blocked). (STEP 1) Check TPMT activity before starting.
Minocycline Tetracycline	Inhibits matrix metalloproteinases .	RA; safe but low potency.	Dizziness/ vestibular toxicity , nausea; rare drug-induced lupus (P-ANCA+), hepatitis.	The tetracycline that causes drug-induced lupus and vestibular symptoms.
Leflunomide also in Inflammation Conf.	Reversibly inhibits dihydroorotate dehydrogenase → ↓ pyrimidine synthesis → ↓ T-cell proliferation.	Methotrexate alternative in RA; PsA.	Hepatotoxicity , diarrhea, hypertension, teratogenic . No marrow suppression.	Very long half-life (enterohepatic) → cholestyramine washout before pregnancy.
Mycophenolate mofetil MMF also in Inflammation Conf.	Reversibly inhibits IMP dehydrogenase → ↓ de novo guanosine synthesis; lymphocytes lack a salvage pathway.	Lupus nephritis induction/maintenance, transplant, DM steroid-sparing (not RA joints).	(STEP 1) GI upset, cytopenias, infection, teratogenic.	Lupus nephritis induction = steroids + MMF or cyclophosphamide .

BIOLOGICS — KNOW BY MECHANISM

DRUG	MECHANISM	USES	TOXICITIES & CONTRAINDICATIONS	HIGH-YIELD PEARL
TNF-α inhibitors Etanercept (decoy receptor + IgG1 Fc) · infliximab (chimeric) · adalimumab, golimumab (human) · certolizumab pegol (PEG Fab) also in Inflammation Conf.	Bind/neutralize TNF-α . "Etanercept intercepts TNF."	RA (+ methotrexate), AS/axial SpA, PsA, psoriasis. Monoclonals also IBD and uveitis — etanercept does not.	TB reactivation , hepatitis B reactivation, serious infection, drug-induced lupus, rare demyelination. Infliximab: anti-drug antibodies, serum sickness (type III).	Screen latent TB (PPD/IGRA) first — TNF maintains granulomas. ~70% reach ACR20. Avoid in lupus (drug-induced lupus).
Anakinra Recombinant IL-1 receptor antagonist · daily SC	Blocks the IL-1 receptor .	Very effective in periodic fever syndromes (CAPS), systemic JIA/Still, acute gout, pericarditis; moderate in RA.	Injection-site welts, neutropenia , infection.	The "-ra" = receptor antagonist; think IL-1-driven autoinflammatory disease.
Rituximab Anti-CD20 also in Inflammation Conf.	Depletes CD20+ B cells (complement- and antibody-dependent killing) for ≥ 6 months.	RA (not PsA/AS), GPA induction/maintenance, refractory lupus, myositis.	Infusion reactions, PML (JC virus) , hepatitis B reactivation (check HBcAb).	New focal deficits + white-matter lesions on rituximab → PML .
Abatacept CTLA4-Ig fusion protein	Binds CD80/86 (B7) on APCs → blocks CD28 co-stimulation of T cells.	RA (IV or SC); PsA.	Well tolerated; slow onset; moderately effective.	"Second signal" blocker: CTLA4-Ig → no CD28-B7 .
Tocilizumab Anti-IL-6 receptor	Blocks the IL-6 receptor .	RA refractory to anti-TNF, systemic JIA, giant cell arteritis , Castleman, COVID-19 inflammation.	Infections, $\uparrow$ BP, $\uparrow$ liver enzymes.	Think GCA . IL-6 drives CRP, so CRP stays low on tocilizumab even with infection.
Belimumab BLyS/BAFF inhibitor also in Inflammation Conf.	Blocks B-lymphocyte stimulator (BAFF), which keeps B cells alive.	Lupus — joints, skin, and appears to help kidneys.	Infection.	First new lupus drug in decades.
Tofacitinib JAK inhibitor · oral small molecule	Blocks JAK-STAT cytokine signaling (lecture: JAK3).	RA; PsA.	Infections (zoster), $\uparrow$ cholesterol .	Congenital JAK3 absence → SCID. The only oral targeted agent on the list.
IL-17 and IL-12/23 inhibitors Taught by target only	Block IL-17 (Th17) or the shared p40 of IL-12/23.	IL-17: ankylosing spondylitis, PsA, psoriasis. IL-12/23: psoriasis skin, PsA.	STEP 1 IL-17 blockade can worsen/unmask IBD; candidiasis.	Axial SpA: NSAIDs → anti-TNF or IL-17 .
OTHER IMMUNOTHERAPIES				
IVIG Pooled human IgG, IV	Replaces antibodies in immunodeficiency; modulates autoimmunity (Fc-receptor blockade).	Kawasaki, ITP, Guillain-Barré, refractory dermatomyositis.	Infusion reactions, thrombosis , renal injury.	Immunodeficiency replacement <i>and</i> autoimmune modulation.
Pegloticase	Converts uric acid → soluble allantoin .	Refractory gout ; dramatically lowers serum urate.	~50% develop neutralizing antibodies ; infusion reactions.	Check serum uric acid: if it isn't low, antibodies have neutralized the drug .

DRUG	MECHANISM	USES	TOXICITIES & CONTRAINDICATIONS	HIGH-YIELD PEARL
PEGylated porcine uricase · IV				

Inflammation Conference

Mechanisms of rheumatologic therapies · approach to cases (drugs not already in the DMARDs lecture)

DRUG	MECHANISM	USES	TOXICITIES & CONTRAINDICATIONS	HIGH-YIELD PEARL
SYMPTOM CONTROL				
NSAIDs e.g., naproxen	Reversibly inhibit COX-1 and COX-2 → ↓ prostaglandins.	First line for axial SpA symptoms, gout flares, PsA; relieve symptoms, do not modify disease.	Gastric ulcer /GI bleed, renal ischemia (afferent arteriole constricts), interstitial nephritis, aplastic anemia.	Morning back pain that improves with activity and NSAIDs → inflammatory (axial SpA), not mechanical.
IMMUNOSUPPRESSION FOR SYSTEMIC DISEASE				
Glucocorticoids e.g., prednisone	Block phospholipase A ₂ and NF-κB → broad anti-inflammatory, immunosuppressive effect.	PMR (prednisone 10–20 mg, dramatic response), lupus flares/nephritis, dermatomyositis, GPA, sarcoidosis, acute gout.	Hyperglycemia/diabetes, cushingoid changes, ↑ lipids, cataracts , osteoporosis , avascular necrosis (hip, knee, shoulder), infection.	Lupus patient on long-term prednisone with groin pain → AVN of the femoral head . Steroids alone do not improve renal survival in lupus nephritis.
Cyclophosphamide Alkylating agent · IV (Euro-Lupus low dose) or PO	Alkylates DNA → kills proliferating lymphocytes.	Severe lupus nephritis induction; GPA induction.	High infection risk, hemorrhagic cystitis and bladder cancer, premature ovarian failure /infertility.	Mesna + hydration protect the bladder (mesna binds the toxic metabolite acrolein in urine). Discuss fertility preservation.
Calcineurin inhibitors Cyclosporine, tacrolimus, voclosporin	STEP 1 Inhibit calcineurin → ↓ NFAT → ↓ IL-2 transcription → ↓ T-cell activation.	Lupus nephritis — reduce proteinuria .	STEP 1 Nephrotoxicity , hypertension; tacrolimus: hyperglycemia, neurotoxicity; cyclosporine: gingival hyperplasia, hirsutism.	Newer lupus-nephritis add-on that lowers proteinuria.
Anifrolumab Anti-type I interferon receptor mAb	Blocks the type I IFN receptor .	Lupus skin and joints; benefit for nephritis unclear.	STEP 1 Infections (respiratory, herpes zoster), infusion reactions.	Lupus is an interferon-signature disease — this drug targets it directly.

Cross-cutting high-yield rules

- **Only anabolics build bone** (teriparatide, abaloparatide, romosozumab). Every anabolic must be **followed by an antiresorptive**. Very-high fracture risk → start anabolic.
- **Drug holiday = bisphosphonates only**. Never stop denosumab without a bisphosphonate → rebound multiple vertebral fractures.
- **Signature risks**: denosumab → hypocalcemia in advanced CKD · raloxifene/estrogen → VTE · romosozumab → MI/stroke · PTH analogs → osteosarcoma-risk patients · bisphosphonates/denosumab → ONJ, atypical femoral fracture.
- **Before any biologic**: screen latent TB (PPD/IGRA) and hepatitis B. TNF maintains granulomas; rituximab reactivates HBV and causes PML.

- **RA ladder:** NSAIDs/prednisone → low-dose weekly methotrexate (+ folic acid) → add a biologic (anti-TNF) after a negative TB test. Treat to target (DAS28, ACR20/50/70, RAPID3).
- **Anti-drug antibodies** cause secondary failure: chimeric drugs and stop-start dosing raise the risk; regular dosing plus methotrexate lowers it.
- **Lupus:** hydroxychloroquine for nearly everyone; nephritis → steroids + mycophenolate or cyclophosphamide; steroids alone do not protect the kidney.

Colchicine is not in the pharmacology lectures, but the rheumatology cases lecture names it for gout: it binds tubulin → blocks microtubules → stops neutrophil chemotaxis; GI, neuromyopathy and marrow toxicity; avoid with renal/hepatic failure and CYP3A4 inhibitors. Psoriasis/spondyloarthritis biologics (IL-17, IL-12/23) are taught by target only, so they appear here without drug names.